

WHEN TO USE WHAT — match your situation · CI = estimate $\pm$ (c.v.)(e.s.e.) · Reject H_0 if p-value $\leq \alpha$

Situation	Test	Statistic	df	Situation	Test	Statistic	df
1 mean, σ unknown, $n < 30$, normal pop.	S22 t-test	$T = (\bar{x} - \mu) / (s/\sqrt{n})$	$n-1$	2 indep. means, $n_1 \geq 30$ & $n_2 \geq 30$ (or σ known)	S24/25 Z	$Z = (\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	—
1 proportion (yes/no)	S23 Z-test	$Z = (\hat{p} - p_0) / \sqrt{(p_0(1-p_0)/n)}$	—	2 indep. means, small n , larger/smaller $s \leq 1.4$	S25 Pooled t	$T = (\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0) / \sqrt{(s_p^2(1/n_1 + 1/n_2))}$	$n_1 + n_2 - 2$
Paired — same subjects, two treatments	S26 Paired t	$T = (\bar{d} - (\mu_D)_0) / (s_D/\sqrt{n})$	$n-1$	2 indep. means, small n , larger/smaller $s > 1.4$	S25 Unpooled t	$T = (\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	formula ↓
2 proportions (yes/no, 2 groups)	S27 Z-test	$Z = (\hat{p}_1 - \hat{p}_2) / \sqrt{(\hat{p}_1(1-\hat{p}_1)(1/n_1 + 1/n_2))}$	—	Regression / Correlation	S28–30	$\hat{Y} = b_0 + b_1 X$ · $b_1 = s_{xy}/s_x^2$ · $b_0 = \bar{y} - b_1 \bar{x}$ · $r = s_{xy}/(s_x \cdot s_y) \rightarrow$ see green card for full formulas	

Single Mean — σ unknown Set 22

$n \geq 30 \rightarrow Z$ · $n < 30$, normal pop $\rightarrow T$ · e.s.e. = $s/\sqrt{n}$

$$T \text{ or } Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$T \sim t_{n-1}$ ($n < 30$, σ unknown) · $Z \sim N(0, 1)$ ($n \geq 30$)

$$\text{CI: } \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

Single Proportion Set 23

$$\hat{p} = x/n \cdot \text{e.s.e. (CI)} = \sqrt{\hat{p}(1-\hat{p})/n}$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0, 1) \text{ test} \rightarrow \text{use } p_0$$

$$\text{CI: } \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$n = \frac{z_{\alpha/2}^2 \cdot \hat{p}(1-\hat{p})}{E^2} [\hat{p} = 0.5 \text{ if unknown}]$$

Paired Data Set 26 · same subjects, two treatments

1. Compute: $D_i = X_i - Y_i$ (e.g. after – before)
2. Find $\bar{d}$ and s_D ← calculator 1-var stats on the D_i values ($\bar{x} = \bar{d}$, $s_x = s_D$)

$$\mathbf{3.} T = \frac{\bar{d} - (\mu_D)_0}{s_D/\sqrt{n}} \sim t_{n-1}$$

Usually $(\mu_D)_0 = 0$ ("no difference"). Non-zero only if claim says "by k " — e.g. "treatment reduces score by 5" $\rightarrow (\mu_D)_0 = 5$.

$$\mathbf{4.} \text{CI: } \bar{d} \pm t_{\alpha/2, n-1} \cdot \frac{s_D}{\sqrt{n}}$$

Two Proportions Set 27

pool $\hat{p}_p$ for test only; unpooled e.s.e. for CI

$$\hat{p}_p = (x_1 + x_2) / (n_1 + n_2) \leftarrow \text{H-test only}$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_p(1-\hat{p}_p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0, 1)$$

$$\text{CI: } (\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Two Independent Means Sets 24–25 · larger s + smaller $s \leq 1.4 \rightarrow$ Pooled · $> 1.4 \rightarrow$ Unpooled

Pooled (equal var, small n) · $\nu = n_1 + n_2 - 2$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$T = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2}$$

$$\text{CI: } (\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Unpooled (unequal var, small n) or Large n / σ known

$$T \text{ or } Z = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

$T \sim t_\nu$ (small n) · $Z \sim N(0, 1)$ (large n / σ known)

$$\nu = \left\lfloor \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\left(\frac{s_1^2/n_1}{n_1-1} + \frac{s_2^2/n_2}{n_2-1}\right)} \right\rfloor$$

$$\text{CI: } (\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

Regression & Correlation Sets 28–30 · $\hat{Y} = b_0 + b_1 x$ · compute s_{xy} , s_x^2 , s_y^2 first

$$s_{xy} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{n-1} \text{ compute } \sum x_i y_i \text{ by hand}$$

$$s_x^2 = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1} \text{ same for } s_y^2$$

$$b_1 = s_{xy}/s_x^2 \cdot b_0 = \bar{y} - b_1 \bar{x} \cdot \text{residual: } e = y - \hat{y}$$

$$\text{SSE} = (n-1) \left(s_y^2 - \frac{s_{xy}^2}{s_x^2} \right) \cdot s_e = \sqrt{\frac{\text{SSE}}{n-2}}$$

$$r = s_{xy}/(s_x \cdot s_y) \cdot R^2 = r^2$$

$$s_{b_1} = \frac{s_e}{\sqrt{(n-1)s_x^2}} \text{ e.s.e. of } b_1$$

T (slope): $T = b_1/s_{b_1} \sim t_{n-2}$ · $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$

T (corr): $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$ · $H_0: \rho = 0$

Critical values (c.v.):

Two-tail: 90% $\rightarrow 1.645$ · 95% $\rightarrow 1.96$ · 99% $\rightarrow 2.576$

One-tail: 90% $\rightarrow 1.28$ · 95% $\rightarrow 1.645$ · 99% $\rightarrow 2.33$

p-value from z-table (table gives $P(Z < z)$):

$H_1: \theta > \text{val} \rightarrow p = 1 - \text{table}(z_{\text{obs}})$ upper tail

$H_1: \theta < \text{val} \rightarrow p = \text{table}(z_{\text{obs}})$ lower tail

$H_1: \theta \neq \text{val} \rightarrow p = 2(1 - \text{table}(|z_{\text{obs}}|))$ two-tail

p-value from t-table (same tail rules, bracket t_{obs}

by df):

e.g. $t_{\text{obs}} = 2.49$, df = 11:

$t_{0.025, 11} = 2.201 < 2.49 < t_{0.01, 11} = 2.718$

$\rightarrow 0.01 < p < 0.025$ (one-tail)

Traps:

CI prop $\rightarrow \hat{p}$; test $\rightarrow p_0$

2-prop test $\rightarrow$ pool $\hat{p}_p$; CI $\rightarrow$ no pool

$(\mu_1 - \mu_2)_0 = 0$ unless claim says "by k "

Before/after $\rightarrow$ paired, not 2-sample